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Full Stack Web Development

Semester Project Report

Pharmacy Management System.

Submitted By:

Muhammad Aarish Mughal – 223570

Aymen Majid – 223574

Kawish Hassan – 223578

Submitted To:

Sir Rashaf Jamil.

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Abstract

The Pharmacy Management System is a full-stack web application developed using the MERN approach. The system enables pharmacists to manage medicines, manufacturers, vendors, customers, and sales records through a clean, user-friendly interface. The project demonstrates CRUD operations, modular component design, and efficient state management with React, while Express.js handles server-side routing and data operations.

Introduction

The goal of this project was to build a web-based application that simplifies the process of managing a pharmacy. The application provides interfaces to manage medicines, customers, vendors, manufacturers, and track medicine sales in real time.

Objectives

- Enable pharmacists to manage inventory with ease
- Provide modular sections for different entities: medicines, vendors, customers, etc.
- Ensure data consistency and quick access via file-based storage
- Create an intuitive frontend for seamless interaction

Technologies Used

Layer	Technology
Frontend	React.js, React Bootstrap
Backend	Node.js, Express.js
Data Storage	JSON files (used in place of MongoDB)
Styling	Bootstrap CSS
Tools	Postman (for API testing), VS Code, GitHub

System Architecture

- **Frontend:** React app with components like MedicineForm.js, CustomerForm.js, SalesForm.js, etc.
- **Backend:** Express.js API with routes for each entity.
- **Storage:** JSON files such as medicines.json, customers.json used for persistent data.

- **Communication:** RESTful API via axios for CRUD operations.

Database Design

Each JSON file is structured to simulate a database-like format.

Example: medicines.json

```
[
  {
    "id": 1,
    "name": "Paracetamol",
    "manufacturer": "ABC Pharma",
    "price": 50,
    "quantity": 100
  }
]
```

Each entity (medicine, vendor, customer, sale) has its own structure and is stored separately.

Backend Implementation

- Located in server.js
- CRUD endpoints for:
 - /api/medicines
 - /api/customers
 - /api/vendors
 - /api/manufacturers
 - /api/sales

Frontend Implementation

- Pages built using React Functional Components
- Routing handled by react-router-dom
- Bootstrap components used for styling
- Pages include:

- Home.js: Project landing page
- Navbar.js: Top navigation
- Forms: MedicineForm.js, CustomerForm.js, SalesForm.js, etc.
- Lists: MedicineList.js, CustomerList.js, etc.

Features Overview

- Add/View/Delete Medicines
- Add/View Customers
- Add/View Vendors
- Add/View Manufacturers
- Add/View Sales
- Responsive UI using Bootstrap
- All features implemented using React state and Express API

Testing

- Manual testing via React frontend
- Postman used to verify API behavior
- Tested for:
 - Form validation
 - Error handling (e.g., file not found)
 - CRUD consistency

Conclusion

This project demonstrates the power and flexibility of the MERN-like stack, even with simplified JSON-based storage. The Pharmacy Management System fulfills the key functional requirements and lays the foundation for a scalable and real-world application with further improvements.